

NETWORKS AND COMPLEXITY

Solution 10-9

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 10.9: Targeted attacks in general [Lvl. 3]

In this exercise we derive some results on degree-targeted attacks in general. Consider a configuration model network, described by a degree generating function G , that is subject to an attack described by a removal function R . Try to do the calculations below in the forward direction (deriving the result) instead of the backward direction (proving the result is true).

- a) Show that the degree distribution of the surviving nodes after the first step of the attack is generated by $G_h = \frac{G-R}{c}$.

Solution

We can start by recalling that the number of nodes with degree k is

$$n_k = Np_k. \quad (1)$$

Moreover, the number of nodes of degree k that are removed in the attack is Nr_k . So the number of nodes of degree k that survive the first step of the attack are

$$n_{k,h} = N(p_k - r_k) \quad (2)$$

We define

$$c = 1 - r = 1 - \sum r_k \quad (3)$$

as the proportion of nodes that survive the attack. Hence the total number of nodes in the network after the attack is

$$N_h = cN \quad (4)$$

We can now compute the proportion of nodes of degree k after the first step of the attack as

$$p_{k,h} = \frac{n_{k,h}}{N_h} = \frac{N(p_k - r_k)}{cN} = \frac{p_k - r_k}{c} \quad (5)$$

Now we multiply both sides of the equation with x^k and sum over all k ,

$$\sum p_{k,h}x^k = \sum \frac{p_k - r_k}{c}x^k \quad (6)$$

Finally, using the definitions of the generating functions $G_h = \sum p_{k,h}x^k$, $G = \sum p_kx^k$, and $R = \sum r_kx^k$, we arrive at the desired result

$$G_h = \frac{G - R}{c} \quad (7)$$

- b) Show that the mean degree after the attack is $z_a = z\tilde{c}^2/c$.

Solution

First we recall that the degree generating function after the attack is

$$G_a = \frac{G(\tilde{A}) - R(\tilde{A})}{c} \quad (8)$$

where

$$\tilde{A} = \tilde{r} + \tilde{c}x, \quad (9)$$

and $\tilde{r}$, $\tilde{c}$ are the removed and surviving proportions of endpoints respectively. They are defined by

$$\tilde{r} = \frac{R'(1)}{z}, \quad \tilde{c} = 1 - \tilde{r} \quad (10)$$

Let's compute z_a as

$$z_a = G'_a(1) = \frac{G'(\tilde{A}(1)) - R'(\tilde{A}(1))}{c} \tilde{A}'(1) = \frac{G'(1) - R'(1)}{c} \tilde{c} = \frac{z - R'(1)}{c} \tilde{c} \quad (11)$$

At this point the $R'(1)$ is bothering us a little bit, so how can we get rid of it? We have last seen it in the definition of $\tilde{r}$ and using this definition we can replace

$$R'(1) = z\tilde{r} \quad (12)$$

and hence

$$z_a = \frac{z - z\tilde{r}}{c} \tilde{c} = \frac{z(1 - \tilde{r})\tilde{c}}{c} = z \frac{\tilde{c}^2}{c}, \quad (13)$$

which is the desired result.

- c) Show that the excess degree distribution after the attack is generated by $Q_a = (G'(\tilde{A}) - R'(\tilde{A}))/\tilde{c}z$.

Solution

We can now compute the excess degree generating function as

$$Q_a = \frac{G'_a}{z_a} = \frac{G'(\tilde{A}) - R'(\tilde{A})}{cz_a} \tilde{A}' = (G'(\tilde{A}) - R'(\tilde{A})) \frac{\tilde{c}}{z\tilde{c}^2} = \frac{G'(\tilde{A}) - R'(\tilde{A})}{z\tilde{c}} \quad (14)$$

- d) Show that the mean excess degree after the attack is $q_a = q - R''(1)/z$.

Solution

Since we already know the generating function Q_a we can compute

$$q_a = Q'_a(1) = \frac{G''(\tilde{A}(1)) - R''(\tilde{A}(1))}{z\tilde{c}} \tilde{A}'(1) = \frac{G''(1) - R''(1)}{z\tilde{c}} \tilde{c} = \frac{G''(1) - R''(1)}{z}. \quad (15)$$

Now we only need to recognize

$$\frac{G''(1)}{z} = q, \quad (16)$$

which leads us to the desired result

$$q_a = q - \frac{R''(1)}{z}. \quad (17)$$